

TABLE III
CLASSIFICATION OF THE INDIAN SECURITY AND SURVEILLANCE MARKET

Market classification	Indian statistics	Major growth drivers
Access control market	According to a report by The Global Security Market, the Indian market for access control was worth US\$ 76 million in 2015 and is projected to nearly double to US\$ 140 million by 2019.	The market for smartcard-based users is a major contributor to the growth of the access control market. The smartcard is used as a medium to authenticate a person's identity and provide access to a person's resources. Physical and logical access convergence improves security by enabling a single smartcard to support multiple authentication methods while helping organisations meet regulatory requirements, enforce consistent policies and drive consolidated audit logs throughout the enterprise.
Video surveillance market	India's video surveillance market is projected to grow at a CAGR of nearly 13 per cent during 2017-2023. It is expected to top US\$ 2.4 billion by 2020.	Even though a majority of market revenues is generated by analogue-based surveillance systems, growing awareness of public safety and falling prices of technology-driven security techniques have proliferated the IP-based surveillance systems.
Intrusion alarm system		Demand for intrusion alarm systems is mainly in the residential sector, hotels, office buildings, etc. Sensor-based security solutions could very soon be moving towards providing business intelligence as well.
System integrated market	The system integration market size is estimated to grow from US\$ 269.62 billion in 2016 to US\$ 387.85 billion by 2021, at a CAGR of 7.5 per cent.	It is driven by the need to automate and integrate the security system, to make it more cost-effective. Advancements in virtualisation and Cloud integration enables benefits such as scalability, no investment in hardware and location independence, which promotes its acceptance. Intrusion alarm systems are security techniques that are designed to detect the presence of intruders at homes or premises. These trigger a local siren to detect suspicious activity and, simultaneously, communicate the event to a security-monitoring centre so that a quick and appropriate response can be carried out.

leverage biometrics to identify and prevent retail crimes. According to National Retail Foundation, organised retail crime costs the retail industry around US\$ 30 billion each year, in addition to petty shoplifting. Retail stores also use video analytics to study the behaviour of customers in a store. Through CCTVs, they are able to recognise the likes and demands of their customers and use video footage to assist with store layout, merchandise positioning, staffing and promotions in order to improve customer experience.

Public sector. One of the biggest users of video surveillance is the public sector. Governments of different nations put the safety and security, and the need to protect their citizens, at a high priority. Such an intricate surveillance of the population is called mass surveillance. It is necessary to fight terrorism, prevent social unrest and protect national security.

Education. One of the faster-growing market segments for security technology is schools and universities. Combining proactive protection for students and teachers with digital security technology makes it possible to optimise safety while remaining within budget. Perimeter surveillance, video surveillance, tracking

suspicious activities and reporting these to the authorities, and monitoring educational activities in and out of class, are some of the many solutions that are a result of surveillance techniques.

Industry. With their wide range of applications and features, security cameras are common in most industries around the world. These applications can range from robbery, vandalism, non-adherence to traffic rules to weather monitoring and more. By combining advanced camera features with sturdy outdoor housings or discreet camera design, IP cameras can be installed in almost any location. This makes these a good fit for monitoring of operational processes and business units, recording in remote corners or recording good-quality video to identify suspicious people.

Indian scenario

According to The Industrial Safety Review, the electronic security market in India would be witnessing tremendous growth at a projected CAGR of nearly 32 per cent during 2017-2023, and is expected to grow to US\$ 2.4 billion by 2020. Growing security concerns and sustained growth of infrastructure in the country has contributed significantly to this number. India's northern region registered the highest

Relevance of security and surveillance systems to business

- Reduction of loss, theft and vandalism
- Improved productivity
- Cost-effectiveness
- Increased sense of security
- Reduced insurance cost
- Improved productivity

revenue in 2016, with the states of Delhi-NCR, Haryana, Punjab, Uttarakhand and Uttar Pradesh being the major contributors. Table III gives the overall growth of the various sectors of the Indian security and surveillance along with their growth drivers.

In the Indian market, focus is now shifting from analogue to IP cameras, which is addressed by hybrid video recorders (HVRs). These HVRs fulfil the need of the current Indian video surveillance market. Also, the need for multi-location monitoring is increasing, due to which video management software (VMS) is in high demand. If we combine the IP and analogue camera markets, the size would be close to ₹ 20 billion in India. It is just like a pyramid, like any other market, which is huge in terms of volume.